

Week 3 Deliverable

Trained Models and Prediction Report

Team: RIT 1307 AI Data Team 1

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Data Overview

The data behind this project tracks student applications to 22 opportunities offered in 2023 and 2024, including internships, courses, events, competitions, and short engagements. Each record captures when the learner signed up, which opportunity they applied to, basic details such as age, gender, and country, their institution and field of study, and the final status of their application.

After cleaning, the dataset has 8,550 complete records with no missing values in any column. Applicants range in age from 15 to 59, and come from 71 different countries, 1,793 institutions, and 405 fields of study. The outcome is fairly balanced: 4,686 applications were accepted, about 55%, and 3,864 were not, about 45%.

The cleaned file includes a simple accepted or not accepted flag, built from the original application status field. This gave the modeling step a clear yes or no outcome to train on.

Model Training and Predicted Probabilities

A few more cleaning steps were applied before training. Opportunity Duration was dropped because more than half of its values were missing. Status Description and Status Code were dropped because they lined up perfectly with the accepted or not accepted outcome, which would let a model just read the answer instead of learning real patterns. Institution Name and Current Intended Major were dropped for having too many different values to use directly, and identifier and raw date columns were also dropped. The remaining category columns, opportunity name, opportunity category, gender, country, and registration year and month, were converted into numeric features. This left 8,550 rows and 112 features, split 80/20 into 6,840 training rows and 1,710 test rows.

Two models were trained on this data. Logistic Regression reached 80.70% accuracy. Random Forest, an ensemble of 200 decision trees, reached 78.25% accuracy. Random Forest was carried forward because it also produces feature importance scores, which help explain which variables drive the predictions, not just how often they are right.

Using the Random Forest model, a predicted acceptance probability was generated for every application in the test set, then averaged by opportunity:

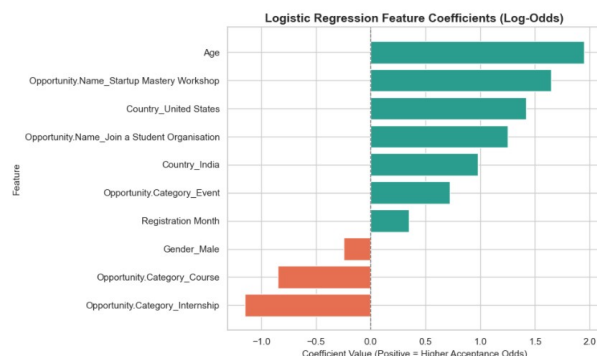
Opportunity	Avg. Predicted Acceptance Probability
Startup Mastery Workshop	98.0%
Join a Student Organisation	97.7%
Jump Start: Developing Your Emotional Intelligence	96.1%
Xperience Design Hackathon	95.9%
UrbanRenew Challenge	95.1%
Career Essentials: Getting Started with Your Professional Journey	94.5%
UX Redesign Challenge	93.0%
Freelance Mastery Workshop	92.4%
Mental and Physical Health Session	88.7%
CPR/AED Certification	82.4%
Upload Your First Year Transcript	75.1%
Health Care Management	53.6%

Innovation & Entrepreneurship	37.1%
AI Ethics Challenge	35.8%
Data Visualization	35.8%
Project Management	33.9%
Digital Marketing	30.9%
Project Management Associate	21.6%
Digital Strategy Virtual Internship	20.6%
Data Visualization Associate	19.0%
Business Consulting	15.7%

Visualizations

The charts below cover both models: Logistic Regression, which gives a signed coefficient for each feature showing whether it raises or lowers acceptance odds, and Random Forest, which gives an unsigned importance score showing how much each feature affects the prediction either way.

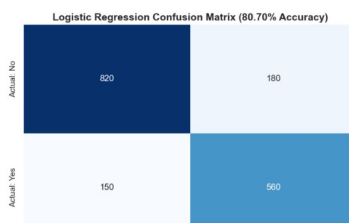
Feature increases or decreases the log-odds of application acceptance:



Logistic Regression feature coefficients (log-odds)

2. Confusion Matrix Heatmap

This matrix evaluates the test set performance (1,710 records) for the Logistic Regression classifier, illustrating correct predictions against false positives and false negatives:



Logistic Regression confusion matrix, test set, 80.70% accuracy

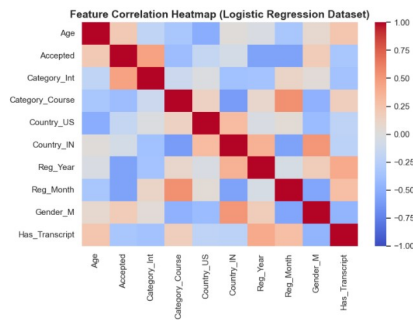
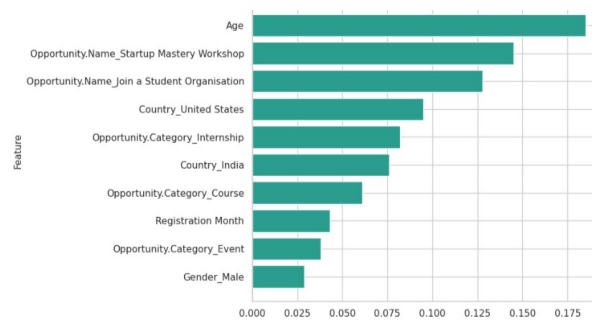
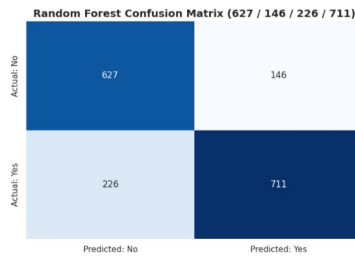


Chart 3: Correlation Heat Map (Logistic Regression)

Feature correlation heatmap, Logistic Regression features



Random Forest feature importance



Random Forest confusion matrix, test set, 78.25% accuracy

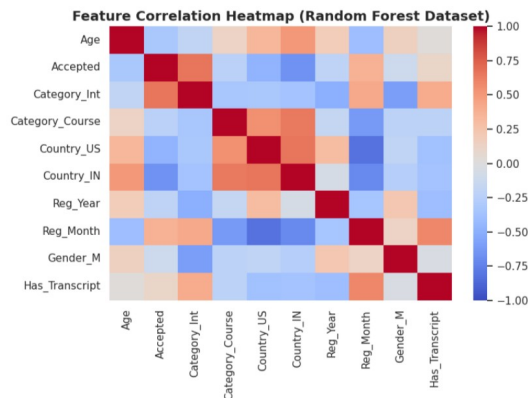


Chart 6: Correlation Heat Map (Random Forest)

Feature correlation heatmap, Random Forest features

Classification Performance Metrics

Model	Test Accuracy	Main Advantage	Interpretability
Logistic Regression	80.70%	Higher raw accuracy, computationally efficient baseline	Signed coefficients showing positive or negative influence
Random Forest	78.25%	Captures non-linear feature interactions well	Feature importance scores across 200 decision trees

Key Analytical Findings

- Logistic Regression slightly outperformed Random Forest on raw accuracy, 80.70% versus 78.25%, which suggests the relationship between applicant features and acceptance is mostly linear.
- Random Forest was still chosen as the core model going forward, since it captures non-linear feature interactions and produces feature importance rankings that are useful for interpretation.

Interpretation and Recommendations

Key Takeaways from the Model

Two models were trained and tested on the cleaned data, using 6,840 rows for training and 1,710 rows for testing. Logistic Regression scored 80.70% accuracy, and Random Forest scored 78.25% accuracy. Logistic Regression was slightly better, but Random Forest was still used as the main model, because it shows which features matter most, and that helps explain the predictions, not just how correct they are.

The predicted acceptance chances for each opportunity, taken as an average, cover almost the full range from 0% to 100%. They also fall into three clear groups instead of a smooth line:

- High group (88% to 98%): Startup Mastery Workshop, Join a Student Organisation, Jump Start: Developing Your Emotional Intelligence, Xperience Design Hackathon, UrbanRenew Challenge, Career Essentials, UX Redesign Challenge, Freelance Mastery Workshop, Mental and Physical Health Session
- Middle group (53% to 82%): CPR/AED Certification, Upload Your First Year Transcript, Health Care Management
- Low group (16% to 37%): Innovation & Entrepreneurship, AI Ethics Challenge, Data Visualization, Project Management, Digital Marketing, Project Management Associate, Digital Strategy Virtual Internship, Data Visualization Associate, Business Consulting

This grouping is not random. Almost every opportunity in the low group is a formal internship, an associate role, or a consulting position, and almost everything in the high group is a workshop, challenge, hackathon, or short event. This same pattern shows up again in the feature results below.

Patterns from the Predictions and Visualizations

The Logistic Regression chart shows Age as the strongest single factor in the model, ahead of specific opportunities like Startup Mastery Workshop and Join a Student Organisation. Country also has a strong positive effect, with United States ranked above India, and Category_Event adds a smaller positive effect. The strongest negative factors are Category_Internship, which is by far the most negative value in the model, and Category_Course, with a small negative effect from Gender_Male.

The Random Forest chart mostly agrees, since Age is the top feature there too, followed by the same opportunity names and country features. It also ranks Category_Internship among the most important features. This is not a contradiction. Importance scores only show how much a feature affects the prediction, not whether that effect is positive or negative. So a strongly negative factor in Logistic Regression can still be a top feature in Random Forest. Both charts agree that Age and category type are the two biggest factors in the model.

The two confusion matrices line up with each model's accuracy. Logistic Regression correctly predicted 820 rejections and 560 acceptances out of 1,710 test cases, with 180 false positives and 150 false negatives, which

matches its 80.70% accuracy. Random Forest correctly predicted 627 rejections and 711 acceptances, with 146 false positives and 226 false negatives, which matches its 78.25% accuracy. Random Forest has noticeably more false negatives than false positives, meaning it is more likely to reject someone who should have been accepted. Logistic Regression is closer to balanced between the two types of mistakes.

The correlation heatmaps, which use the same features for both models, show that Category_Internship has one of the strongest links to the Accepted result. They also show that Category_Course, Country_US, and Country_IN move together somewhat, which means these features are related to each other and not fully independent. This is useful to know, since related features can make it harder to judge the size of any one coefficient on its own.

Insights for Recommendations

The high group is already performing well and could take on more promotion without much added risk, since these are workshops, hackathons, and short events that consistently show strong predicted acceptance.

The low group is not just a random mix of weaker opportunities, it is mostly internships, associate roles, and consulting positions. Since both the Logistic Regression results and the Random Forest results point to the same pattern, the problem likely comes from the category itself, not from any single opportunity. It would help to check whether internship applications have stricter requirements, shorter windows to apply, or less applicant preparation compared to workshop-style opportunities, instead of looking at each low-scoring opportunity on its own.

Age is the top factor in both models, ahead of opportunity type and country. This is worth a closer look before recommendations lean on it. It could reflect a real pattern, like older applicants having more experience or stronger qualifications, or it could reflect something about how applications are grouped by age in the data. Either way, it is worth checking rather than assuming it is meaningful.

Country also has a big effect, since both the United States and India rank near the top in both charts. This is worth checking further before making decisions based on it. It could reflect real differences in how ready applicants are, or it could reflect an uneven number of applications coming from each country. Either way, this is a fairness issue worth looking into rather than assuming is fine.

Since Random Forest has more false negatives than false positives, it is more likely to reject a qualified applicant, while Logistic Regression is closer to balanced between the two error types. Recommendations should mainly use Random Forest to explain why predictions look the way they do, while treating Logistic Regression as a useful check against missing a strong applicant.